

New Physics Predictions

What GTE Predicts Will and Won't Be Found

novaspivack.com/research P48 complete theory monograph

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UGP Physics Tutorial Series

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UGP Physics Tutorial Series

Prerequisites (read first): *How the Standard Model Quantum Numbers Emerge · Forces, Interactions, and the $V-A$ Structure · The Φ_{MDL} Field: From Discrete Certificate to Continuum Physics*

Next tutorials: *Falsifiability: How to Test the GTE Framework*

See also: *The GTE Polynomial: A Step-by-Step Cheat Sheet · The Levels of the Theory*

Claim-Strength Taxonomy

Throughout this tutorial, results are labeled with a confidence grade:

- **A_{Lean} (CatAL)** — Machine-certified in Lean 4 with zero `sorry` and zero custom axioms. Independently verifiable by anyone with Lean installed.
- **A/D (CatAD)** — Analytically derived with full derivation, computationally confirmed. Not yet machine-certified.
- **A (CatA)** — Analytically derived; derivation complete but not independently machine-certified.
- **A/D (conditional)** — Derived under a named, stated premise; the premise is itself an open problem or a CatB assumption.
- **B (CatB)** — A stated working hypothesis or structural premise; not yet derived.

1 Why New Physics Predictions Matter

The core argument

With zero free parameters, every numerical output of the GTE framework is a *genuine prediction* that the framework could have failed but did not. The same zero-parameter structure that derives the Standard Model’s known physics also constrains what lies beyond it. This tutorial catalogs what GTE forbids, what it predicts, and why those predictions are accessible to current and near-future experiments.

Since the Large Hadron Collider began operation, it has confirmed the Higgs boson but has found no evidence of physics beyond the Standard Model (BSM): no supersymmetric particles, no extra dimensions, no new forces. Many BSM theories predicted new particles that should already have appeared. They have not.

GTE (Generative Triple Evolution) is designed differently. Rather than adding new ingredients to the Standard Model, it derives the SM from a single 19-bit algebraic certificate: the polynomial $p(L, C, R) = C + R - CR - LCR$ over $\text{GF}(7)$. This certificate has no free parameters. Its zero-parameter architecture means that any structure not forced by the certificate is absent — and that is precisely why GTE predicts the LHC has found nothing. The theory explains the null result.

But GTE also makes specific predictions about what *will* be found. These predictions are sharp: each has a named experiment, a numerical threshold, and an explicit falsification criterion. This tutorial presents each one with its physical mechanism, its numerical value, and what observation would confirm or refute it.

2 What GTE Forbids: The Structural Absences

2.1 No Supersymmetry

Supersymmetry (SUSY) introduces partner particles (squarks, gluinos, neutralinos) for every SM particle. In GTE, the complete particle spectrum is determined by the $\mathbb{Z}_7$ winding sectors of the Φ_{MDL} field. There are exactly five PSC-admissible winding sectors in the SM branch ($w \in \{0, 2, 3, 4, 6\}$), plus the dark-mirror branch ($w \in \{1, 5\}$). No additional sectors are available. Supersymmetric partners would require a doubling of the winding spectrum with no mechanism to generate it in the GTE certificate structure.

GTE does not forbid SUSY by a theorem that explicitly names SUSY; rather, the complete particle ontology is derived from $\mathbb{Z}_7$ arithmetic and admits no room for superpartners [1]. Every null SUSY result from the LHC is consistent with this prediction.

2.2 No Axion

The **axion** is a hypothetical particle proposed by Peccei and Quinn to explain why the strong CP angle θ_{QCD} is observed to be less than 10^{-10} despite the SM having no reason to favor $\theta = 0$.

In GTE, the strong CP problem is *structurally resolved* without any axion. The internal symmetry group $F_{21} = \mathbb{Z}_7 \rtimes \mathbb{Z}_3$ has abelianisation $F_{21}^{ab} = \mathbb{Z}_3$ — pure color, zero CP-odd phase. This group-theoretic fact forces $\theta_{\text{QCD}} = 0$ exactly, with three independent proofs, all machine-certified (**CatAL**) [1].

Strong CP Prediction (**CatAL**, three independent proofs)

$\theta_{\text{QCD}} = 0$ exactly from F_{21} group theory. No axion is predicted and no axion mechanism is needed. Every null axion search (ADMX, CASPER, BabyIAXO, HAYSTAC) is therefore a **positive confirmation** of the GTE structure, not a failure to find something.

Falsification: A definitive axion detection at any mass and any coupling strength would falsify the GTE strong-CP mechanism.

2.3 No Large Extra Dimensions

Theories with large extra dimensions (e.g., Arkani-Hamed, Dimopoulos, Dvali) explain the hierarchy problem by diluting gravity into higher-dimensional space.

In GTE, the 3+1-dimensional spacetime structure is not postulated — it is derived. The CMCA (**Chiral Minkowski Cellular Automaton**) has exactly one non-trivial persistent-glider sector, which lives in 1+1 dimensions. The continuum Φ_{MDL} field then lifts this to 3+1 dimensions via the Algebraic Lifting Theorem (**CatAL**). A 2+1-dimensional persistent-glider sector analogous to the SM is absent by theorem (the “dimensional staircase” result: no canonical 2+1D discrete substrate compatible with the SM orbit exists; **CatAD**) [1]. No room is available for compact or large extra dimensions within the framework.

2.4 No New Light Particles Below Λ_{GTE}

The GTE certificate structure defines a natural energy scale:

$$\Lambda_{\text{GTE}} = 7 \times M_{\text{kink}} = \frac{8}{7} m_\tau \approx 2.03 \text{ GeV}. \quad (1)$$

Below Λ_{GTE} , the EFT (**effective field theory**) description is complete: the $\mathbb{Z}_7$ winding sectors are all accounted for, and no new stable structures can appear in the SM branch. Any new light stable particle below Λ_{GTE} that is not in the SM spectrum would require a new winding sector, which the certificate structure does not permit.

This does *not* mean the EFT is trivial above Λ_{GTE} . The kink dissolution threshold at Λ_{GTE} is a new-physics scale with specific signatures (see §6).

3 What GTE Predicts: The Affirmative Case

3.1 The Mirror Branch Dark Sector

The $\mathbb{Z}_7$ arithmetic that generates the SM winding sectors $\{0, 2, 3, 4, 6\}$ does not exhaust $\mathbb{Z}_7$. The complementary winding sectors $\{1, 5\}$ are as consistent with $\mathbb{Z}_7$ arithmetic as any SM sector. They form the **mirror branch** — a dark sector forced by the same arithmetic that generates the SM [1].

Definition: Mirror Branch (**CatAD**)

The *mirror branch* consists of $\mathbb{Z}_7$ -admissible three-tape configurations with primary winding number $w = 5$ (the $\mathbb{Z}_7$ conjugate of the SM up-type-quark sector, $w = 2$). Mirror particles carry $\text{SU}(3)_{\text{dark}}$ colour and **no SM gauge charges**: no electric charge, no SM weak charge, no SM color. They are dark by construction — the winding algebra forbids SM-gauge coupling at dimension-4.

The mirror branch has three generations, for the same combinatorial reason that the SM has three: the PSC enumeration that forces $N_{\text{gen}} = 3$ applies to any $\mathbb{Z}_7$ -admissible sector.

3.2 The Mirror-Branch Mass Spectrum

The mass spectrum of the mirror branch is computed by applying the same cascade formula that gives SM lepton masses, with mirror-branch winding inputs. The three mirror lepton masses are [1]:

$$m_{\chi_1} = 0.54 \text{ MeV} \quad (\text{Generation 1, primary dark matter candidate}), \quad (2)$$

$$m_{\chi_2} = 24.5 \text{ MeV} \quad (\text{Generation 2}), \quad (3)$$

$$m_{\chi_3} = 3.60 \text{ GeV} \quad (\text{Generation 3}). \quad (4)$$

These values are **CatAD**. The generation-2/1 mass ratio is $m_{\chi_2}/m_{\chi_1} = 45.4$ — a prediction: if a dark lepton signal is found at any one of these masses, the theory predicts where to find the other two.

The dark sector has a gauge group $\text{SU}(3)_{\text{dark}}$, forced by the same Group of Elegance mechanism that selects QCD. There is *no dark photon*: the absence of a $\text{U}(1)$ factor in the Elegant Kernel for the mirror winding sector is Lean-certified.

No Dark Photon (Lean-certified)

GTE predicts no dark photon at any mass. Every null result from dark-photon searches (NA64, BaBar, Belle II kinematic scan, NA62) is a **positive confirmation** of the mirror-branch structure. Any dark photon discovery at any mass would falsify the GTE dark sector model.

3.3 The Dark Matter Candidate: χ_1 at 0.54 MeV

The lightest mirror lepton χ_1 at 0.54 MeV is the primary dark matter candidate. Its spin-independent cross-section with SM nucleons is predicted to be $\sigma_{\text{SI}} \sim 10^{-61} \text{ cm}^2$ (**CatAD**) [1] — approximately 30 orders of magnitude below any current or projected direct-detection sensitivity. This is not a tuned value; it follows from the winding isolation at dimension-4.

This dark matter is **ultra-faint interacting** (UFIDM): it interacts gravitationally and via $SU(3)_{\text{dark}}$, but is invisible to WIMP-style searches. The predicted relic density is $\Omega_{\text{DM}} h^2 = 0.11994$ (**CatAD**), matching the Planck 2018 measurement to -0.15% without any free parameters.

4 GTE-P7: The Highest-Priority New-Physics Signal

GTE-P7 Prediction (**CatAD**) — Belle II Target

A dark lepton resonance at $m_{\text{GTE-P7}} = 211.9$ MeV is predicted by the mirror-branch winding sector of the GTE polynomial [1]. This is the highest-priority new-physics prediction accessible to running experiments.

4.1 What GTE-P7 Is

GTE-P7 is not χ_1 , χ_2 , or χ_3 , but rather a resonance appearing in the Elegant Kernel computation for the mirror branch. Its mass arises from the same UCL (Universal Calibration Law) pipeline that generates the SM lepton masses, applied to a specific mirror-branch cascade triple.

The numerical value 211.9 MeV is **CatAD**: the derivation chain is complete through the mirror-branch winding sector.

4.2 Where to Look

The GTE-P7 mass lies in the range accessible to Belle II [1], currently taking data, in the channels:

- $e^+e^- \rightarrow \chi? + \text{invisible}$ — a single charged track with missing energy, characteristic of a dark lepton coupled via a portal.
- Displaced-vertex signatures — if the dark lepton decays with a centimeter-scale decay length inside the detector.

The full Belle II dataset (50 ab^{-1} , projected by 2031) will cover the signal window $[200, 225]$ MeV with sufficient sensitivity to confirm or exclude.

Falsification: The GTE-P7 prediction is falsified if no resonance is observed in the 200–225 MeV window after the full 50 ab^{-1} dataset, at the expected coupling scale.

4.3 Why This Mass and Not Another

The cascade formula that places GTE-P7 at 211.9 MeV has no free parameters once the mirror-branch winding sector is specified. The same pipeline that correctly derives m_e , m_μ , m_τ at 0.295% RMS is applied to the mirror-branch triple. The mass prediction is a corollary of the same arithmetic that governs the SM masses. A different mass would require either a different winding sector or different cascade arithmetic — neither of which the framework permits.

5 Proton Stability: Dimension-4 Absolutely Forbidden

5.1 What the SM and GUTs Predict

Grand Unified Theories predict proton decay via the channel $p \rightarrow e^+ \pi^0$, a dimension-6 operator suppressed by M_{GUT}^{-2} , giving a proton lifetime $\tau_p \sim 10^{34} \text{--} 10^{36}$ yr. This is the primary target of Hyper-Kamiokande (under construction) and DUNE.

5.2 The GTE Result: Dimension-4 Forbidden, Dimension-6 Not

In GTE, baryon number is a *topological charge* of the $\mathbb{Z}_7$ winding field:

$$B = \frac{1}{3} \sum_j \chi_q(w_j), \quad (5)$$

where the sum is over the three constituent kink windings. Topological charges are exactly conserved: they cannot change under any local deformation of the field.

This makes dimension-4 baryon-number violation not merely suppressed but *structurally impossible* in GTE. Machine-certified: `gte_baryon_number_topological_charge` (**CatAL**, zero sorry, `Universality/GUTStructure.lean`) [1].

Proton Stability Theorem (**CatAL**)

Dimension-4 proton decay is structurally forbidden: baryon number is a topological winding invariant that cannot be violated by any local operator of dimension ≤ 4 . The proton is topologically stable against dimension-4 operators.

Note: Dimension-6 operators (GUT-scale suppressed, $\sim M_{\text{GUT}}^{-2}$) are *not* forbidden by this theorem. The standard GUT-predicted channel $p \rightarrow e^+ \pi^0$ may exist at the dimension-6 level.

This is one of GTE’s sharpest predictions. A single verified proton decay event identified as a dimension-4 process would falsify the framework. Current experimental limits already place the proton lifetime at $\tau_p > 1.6 \times 10^{34}$ yr for the dominant $e^+ \pi^0$ channel (Super-K).

What the Lean proof establishes: The theorem `gte_baryon_number_topological_charge` certifies that B is conserved as a topological charge of the $\mathbb{Z}_7$ winding field. This means no perturbative operator, no SM radiative correction, and no GUT-scale coupling can violate B at dimension-4 in the GTE EFT.

6 The Λ_{GTE} Scale as a Collider Target

The kink dissolution threshold

$$\Lambda_{\text{GTE}} = 7 M_{\text{kink}} = \frac{8}{7} m_\tau \approx 2.03 \text{ GeV} \quad (6)$$

is the energy at which the GTE EFT description starts to break down: above this scale, individual kinks dissolve into the Φ_{MDL} substrate and new topological structure can form.

What to look for at Λ_{GTE}

- A change in the effective number of degrees of freedom in hadronic cross-sections near $\sqrt{s} \approx 2 \text{ GeV}$.
- Anomalous energy deposition patterns characteristic of kink dissolution in the Φ_{MDL} field.
- Lattice QCD calculations showing a transition in the kink structure function around Λ_{GTE} .

If no structure appears: The absence of any new hadronic feature below $\sim 3 \text{ GeV}$ would be informative but not immediately falsifying, since the transition may be gradual.

The Λ_{GTE} scale is first-principles derived, not a tunable cutoff. It is fixed by m_τ and $|\mathbb{Z}_7| = 7$, both of which are themselves derived. This makes it a genuine prediction, not a parameter.

7 Gravitational Wave Predictions

7.1 Primordial Gravitational Waves: $r = 0$ Exactly

The GTE inflationary mechanism produces scalar perturbations with spectral tilt $n_s = 1 - \ln 2 / (2\pi^2) = 0.96488$ (**CatAL**), confirmed at -0.005σ by Planck 2018 [2].

For the tensor-to-scalar ratio r , GTE predicts:

$$r = 0 \quad (\text{exactly, CatA}). \quad (7)$$

Primordial gravitational waves are absent because the inflationary mechanism in GTE does not involve an inflaton field with large-field excursions — the perturbation spectrum is generated by $\mathbb{Z}_7$ defect formation, not by a slow-roll inflaton.

Current sensitivity: Planck+BICEP/Keck place $r < 0.036$ (95% CL). CMB-S4 (projected ~ 2028) will reach $\sigma(r) \approx 0.002$. A detection of $r > 0.01$ at $> 3\sigma$ would falsify the GTE inflationary mechanism.

7.2 The $\mathbb{Z}_7$ Phase Transition Gravitational Wave Signal

The $\mathbb{Z}_7$ symmetry breaking at temperature $T_G \approx 0.7$ GeV (the domain-wall ordering crossover) produces a network of domain walls that annihilates by $T_{\text{ann}} \approx T_G$ (wall lifetime $\sim 5 \times 10^{-24}$ s). This rapid annihilation generates a gravitational wave background peaked near the frequency corresponding to T_G [1].

GW Background from $\mathbb{Z}_7$ Phase Transition (**CatAD**)

A gravitational wave background from the $\mathbb{Z}_7$ domain-wall network is predicted in the frequency band accessible to the Einstein Telescope and LISA. The zero-relic prediction (complete annihilation before BBN) is a structural GTE result; any detection of a wall relic density above the T_G level would falsify the annihilation mechanism.

8 The Zero-Free-Parameter Principle: Every Output Is a Prediction

The key insight about falsifiability

In a theory with free parameters, a measurement that disagrees with a prediction can always be accommodated by retuning the parameters. In GTE, there are no free parameters to retune. Every output — the nine charged-fermion masses, the Weinberg angle, the dark matter mass, the proton lifetime, r , n_s — is derived, not fitted. A single confirmed disagreement at $> 3\sigma$ precision would refute the theory as currently formulated, with no escape route.

This makes GTE *more* falsifiable than the Standard Model, not less. The SM has 19 free parameters (plus at least 3 for neutrinos) that can absorb any discrepancy. GTE has zero. The absence of new LHC physics is not a negative result for GTE — it is a confirmed prediction.

The spirit of GTE predictions: Every new measurement by Belle II, DUNE, CMB-S4, Hyper-Kamiokande, the Einstein Telescope, or Euclid is simultaneously a test of GTE. The framework does not hide in parameter space; it makes a commitment at each observable. The complete falsification register is presented in the *Falsifiability* tutorial [3].

Table 1: New-physics predictions and structural absences of GTE. “Absence” means structurally forbidden by the certificate; “Prediction” means explicitly derived.

Item	GTE position	Key experiment
Supersymmetric particles	Absent (no winding sector)	LHC (confirmed null)
Axion	Absent ($\theta_{\text{QCD}} = 0$ from F_{21})	ADMX, CASPER (confirmed null)
Dark photon	Absent (no U(1) in dark sector)	NA64, Belle II (confirmed null)
Large extra dimensions	Absent (3+1D forced)	LHC (confirmed null)
GTE-P7 at 211.9 MeV	Predicted (CatAD)	Belle II
χ_1 dark matter at 0.54 MeV	Predicted (CatAD)	Sub-MeV DM experiments
χ_3 at 3.60 GeV	Predicted (CatAD)	LHC mono-jet, Higgs invisible
Dim-4 proton decay: none	Predicted (CatAL)	Hyper-K, DUNE
$r = 0$ primordial tensors	Predicted (CatA)	CMB-S4 (~ 2028)
$\mathbb{Z}_7$ GW background	Predicted (CatAD)	Einstein Telescope
$\Omega_{\text{DM}} h^2 = 0.11994$	Predicted (CatAD)	Planck (-0.15%)
$\Sigma m_\nu = 59.4$ meV	Predicted (CatA)	CMB-S4, Euclid

9 Summary: The Complete New-Physics Picture

The new-physics picture from GTE is internally consistent and makes a clear experimental commitment: a dark sector with specific masses (the mirror branch), proton stability at dimension-4, no axion, no SUSY, and a specific GW background. Each prediction carries a named falsification criterion.

Further Reading

Primary Sources

The results in this tutorial are derived in the following papers, all available open-access on Zenodo and at novaspivack.com/research:

- **P48 — The Complete GTE Framework** (main synthesis monograph):
doi.org/10.5281/zenodo.20560550
- **P47 — Cosmological Predictions of the GTE/ Φ_{MDL} Framework** (cosmological predictions including n_s , r , and the GW background):
doi.org/10.5281/zenodo.20465803
- **P08 — From UGP to GTE: Prime-Locked Universes** (mirror-branch dark sector derivation):
doi.org/10.5281/zenodo.20171002
- **P28 — Computational Universality and the Standard Model** (no-CA-replica theorem, dimensional derivation):
doi.org/10.5281/zenodo.20259513

Lean 4 machine proofs: github.com/novaspivack/ugp-lean

References

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